COMPASS Synoptic: Porewater SO4/Cl

September 2026 Samples

2026-11-06

Contents

1 Run Information **2**

1.1 Assess Standard Curves . . . 2

1.2 Assess Check Standards . . . 4

1.3 Assess Blanks . . . 6

1.4 Assess Duplicates . . . 7

1.5 Calculate mmol/L concentrations & salinity, add dilutions . . . 8

1.6 Assess SO4 Spikes . . . 8

1.7 Check if samples are within the range of the standard curve . . . 9

1.8 Check to see if samples run match metadata & merge info . . . 9

1 Run Information

Run date: 2026-11-03
Run by: Zoe Read
Run notes: One Cl check standard and one duplicate were off, unsure why. Some samples were adl but only very slightly over the standard curve (20.1-22 versus 20 as the highest standard).

SO4 file used: COMPASS_Synoptic_CB_MonMon_202609_SO4.txt

Cl file used: COMPASS_Synoptic_CB_MonMon_202609_Cl.txt

Changed sample names:

sample_ID	New_IDs
NA	NA

Samples removed:

```
## [1] NA
```

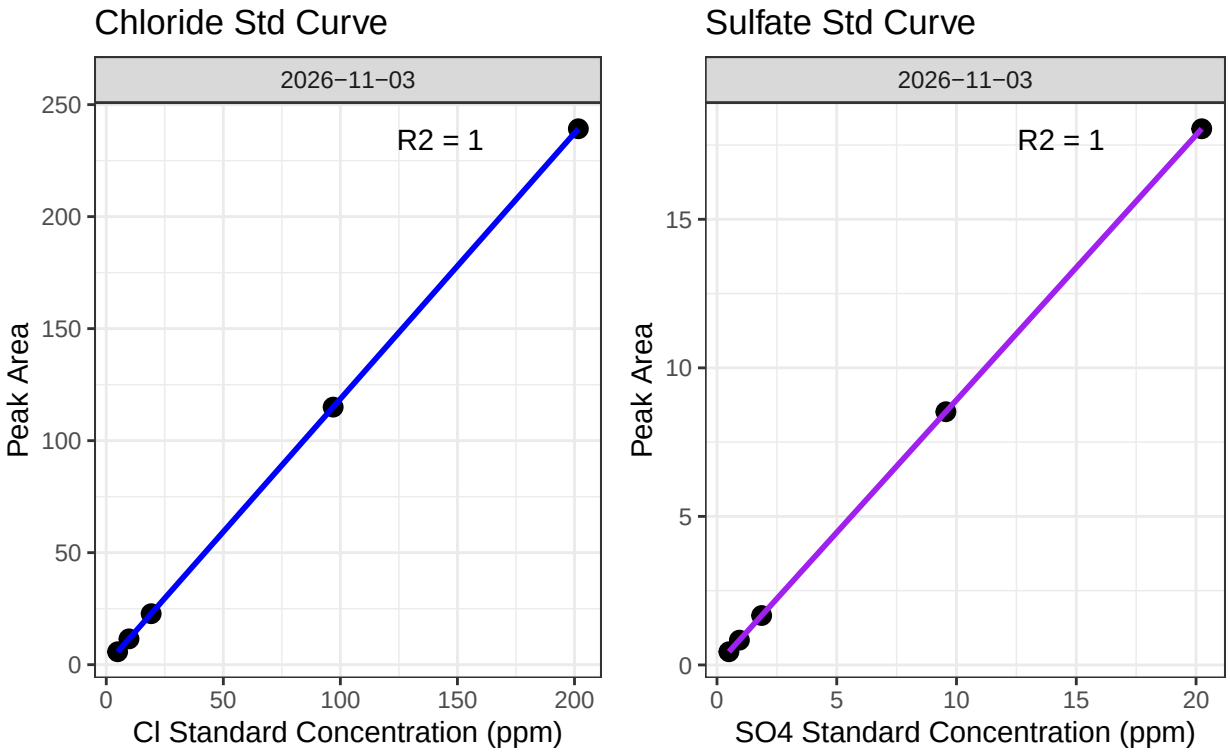
Links:

Base QAQC Code Template Used

Project-Specific QAQC Code Template Used

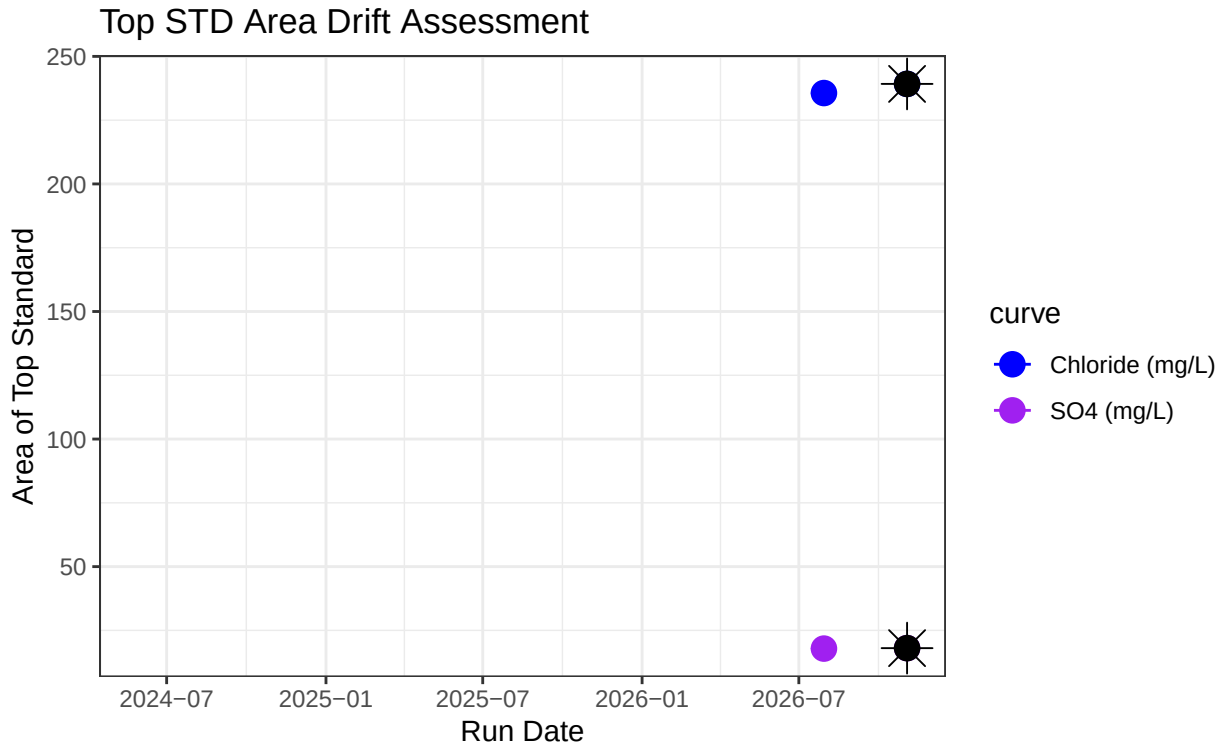
Protocol

1.1 Assess Standard Curves



```
## Continue, overwriting previous standards data for this date?  (Yes/no/cancel)
```






```
## [1] "PROCEED - Cl Curve r2 GOOD"
```

```
## [1] "PROCEED - SO4 Curve r2 GOOD"
```

run_date	analyte	slope_flag	intercept_flag
2026-11-03	Cl	PROCEED - Slope is within 2 sd of previous slopes	PROCEED - Intercept is within 2 sd of previous intercepts
2026-11-03	SO4	PROCEED - Slope is within 2 sd of previous slopes	PROCEED - Intercept is within 2 sd of previous intercepts

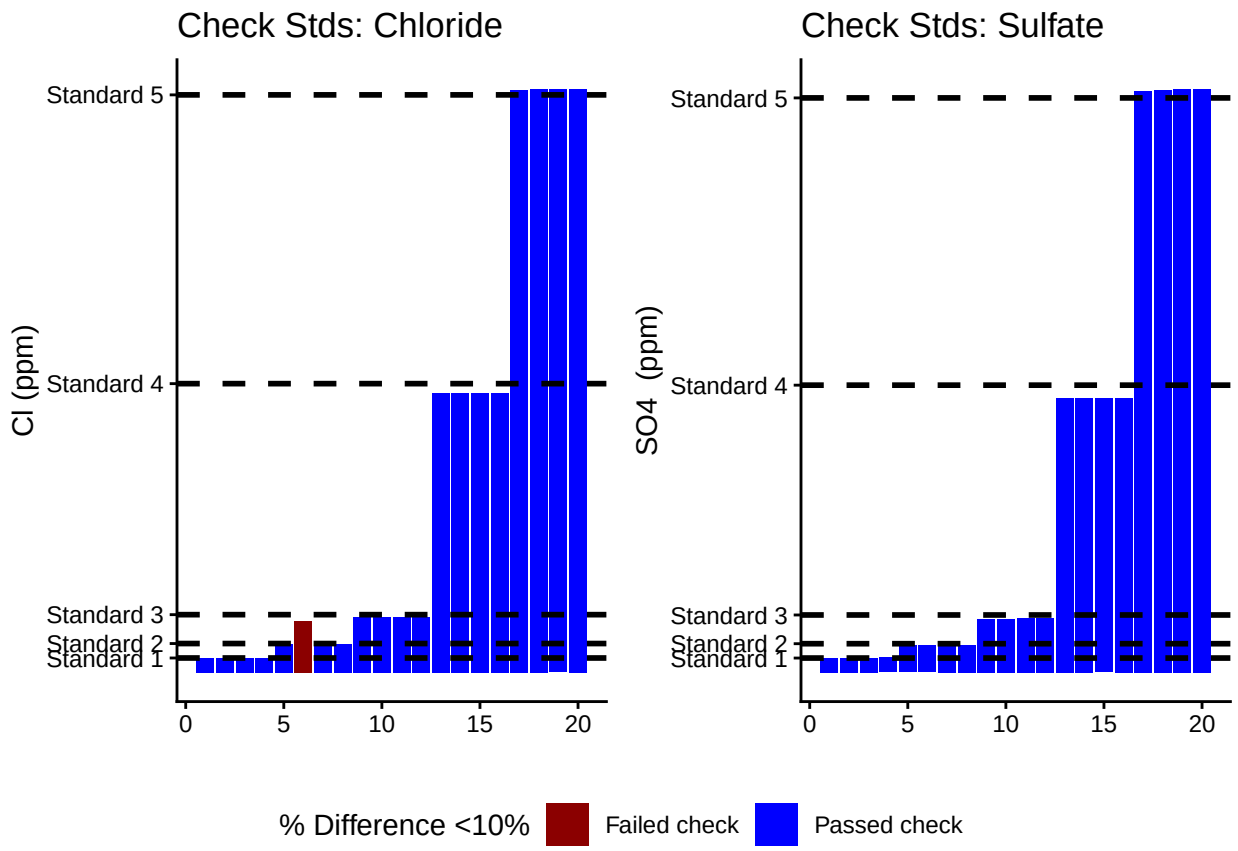
1.2 Assess Check Standards

sample_ID	mean_Cl	cv_Cl	flag_Cl
Standard 1	4.888	0.004	PROCEED - Chloride Check Standard RSD within Range
Standard 2	11.772	0.343	PROCEED - Chloride Check Standard RSD within Range
Standard 3	19.196	0.001	PROCEED - Chloride Check Standard RSD within Range
Standard 4	96.833	0.001	PROCEED - Chloride Check Standard RSD within Range
Standard 5	201.897	0.001	PROCEED - Chloride Check Standard RSD within Range

sample_ID	mean_SO4	cv_SO4	flag_SO4
Standard 1	0.511	0.020	PROCEED - Sulfate Check Standard RSD within Range
Standard 2	0.946	0.013	PROCEED - Sulfate Check Standard RSD within Range
Standard 3	1.877	0.005	PROCEED - Sulfate Check Standard RSD within Range
Standard 4	9.549	0.001	PROCEED - Sulfate Check Standard RSD within Range
Standard 5	20.284	0.002	PROCEED - Sulfate Check Standard RSD within Range

```
## [1] "PROCEED - >80% of Chloride Check Standards have RSD within range"
```

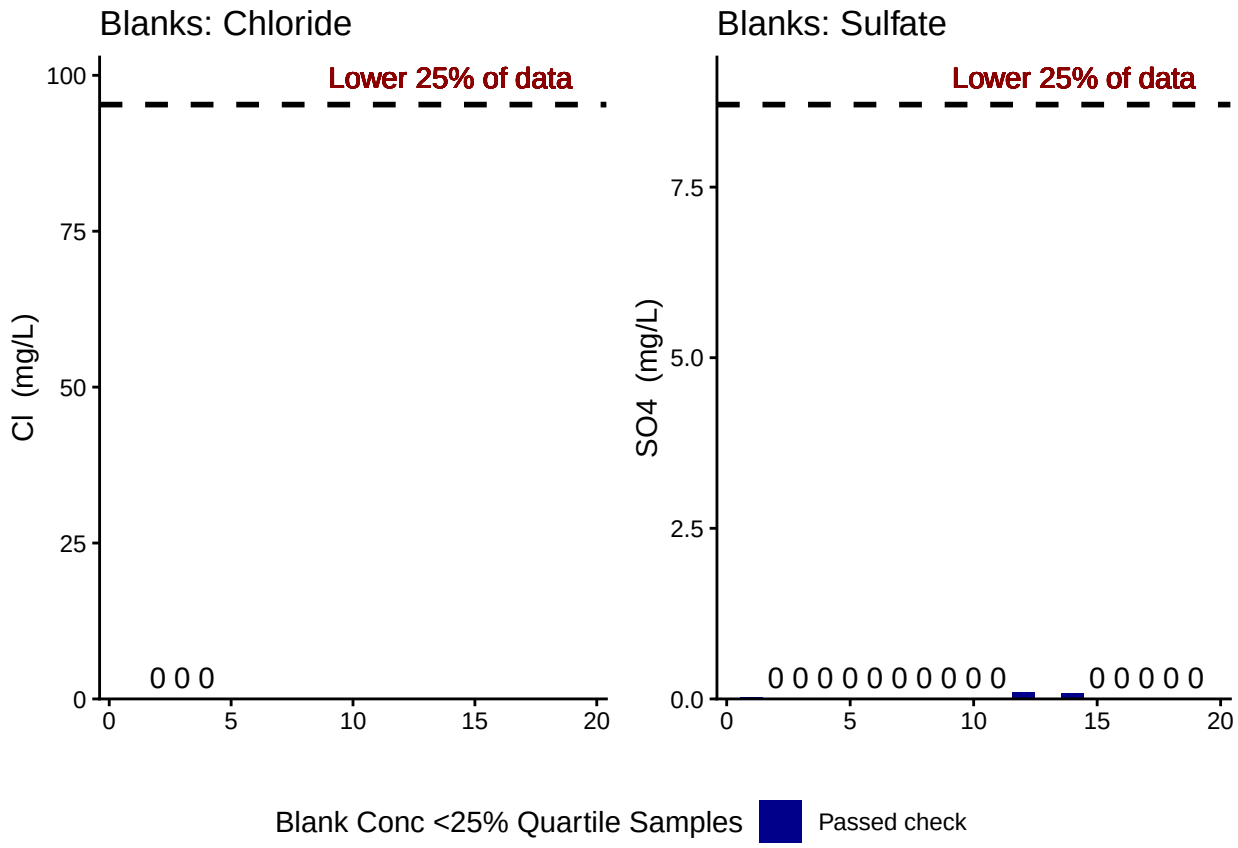
```
## [1] "PROCEED - >80% of Sulfate Check Standards have RSD within range"
```




```
## [1] "PROCEED - >80% of Chloride Check Standards are within range of expected concentration"
```

```
## [1] "PROCEED - >80% of Sulfate Check Standards are within range of expected concentration"
```

1.3 Assess Blanks

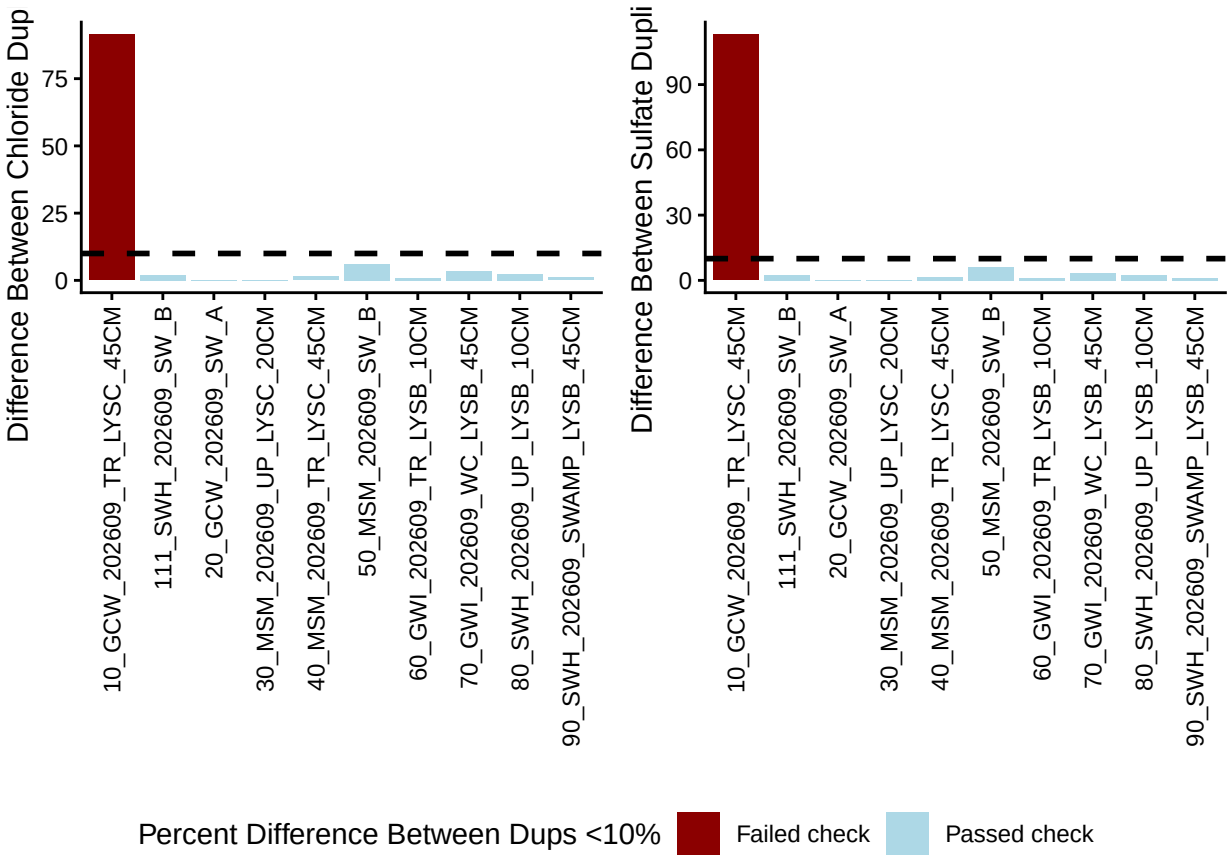


analyte	run_date	mean_blanks
Cl	2026-11-03	0.027
SO4	2026-11-03	0.011

```
## [1] "PROCEED - >80% of Chloride Blank concentrations are in the lower 25% quartile of samples"
```

```
## [1] "PROCEED - >80% of Sulfate Blank concentrations are in the lower 25% quartile of samples"
```

1.4 Assess Duplicates

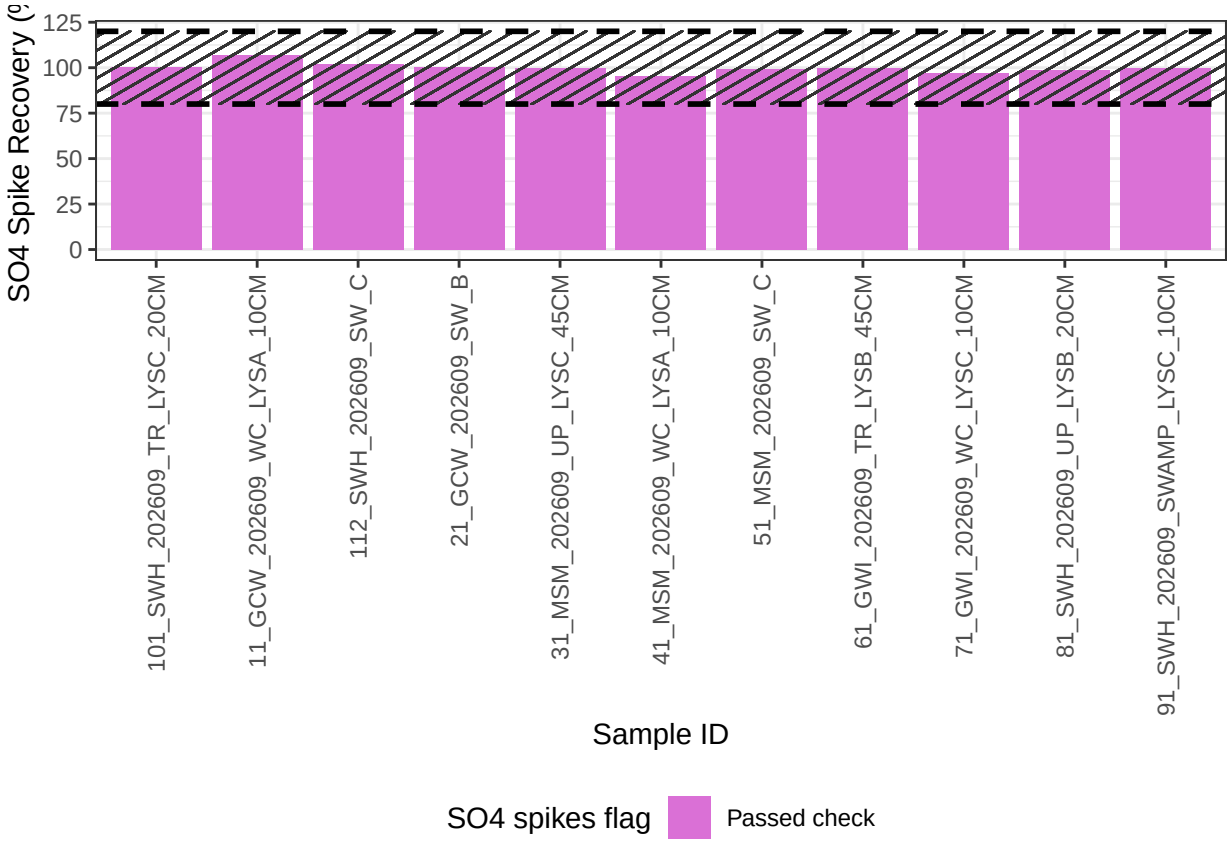



```
## [1] "PROCEED - >80% of Chloride Duplicates have a percent difference <10%"
```

```
## [1] "PROCEED - >80% of Sulfate Duplicates have a percent difference <10%"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

1.6 Assess SO4 Spikes




```
## [1] "PROCEED - >80% of SO4 spikes have a recovery between the high and low cutoff"
```

1.7 Check if samples are within the range of the standard curve

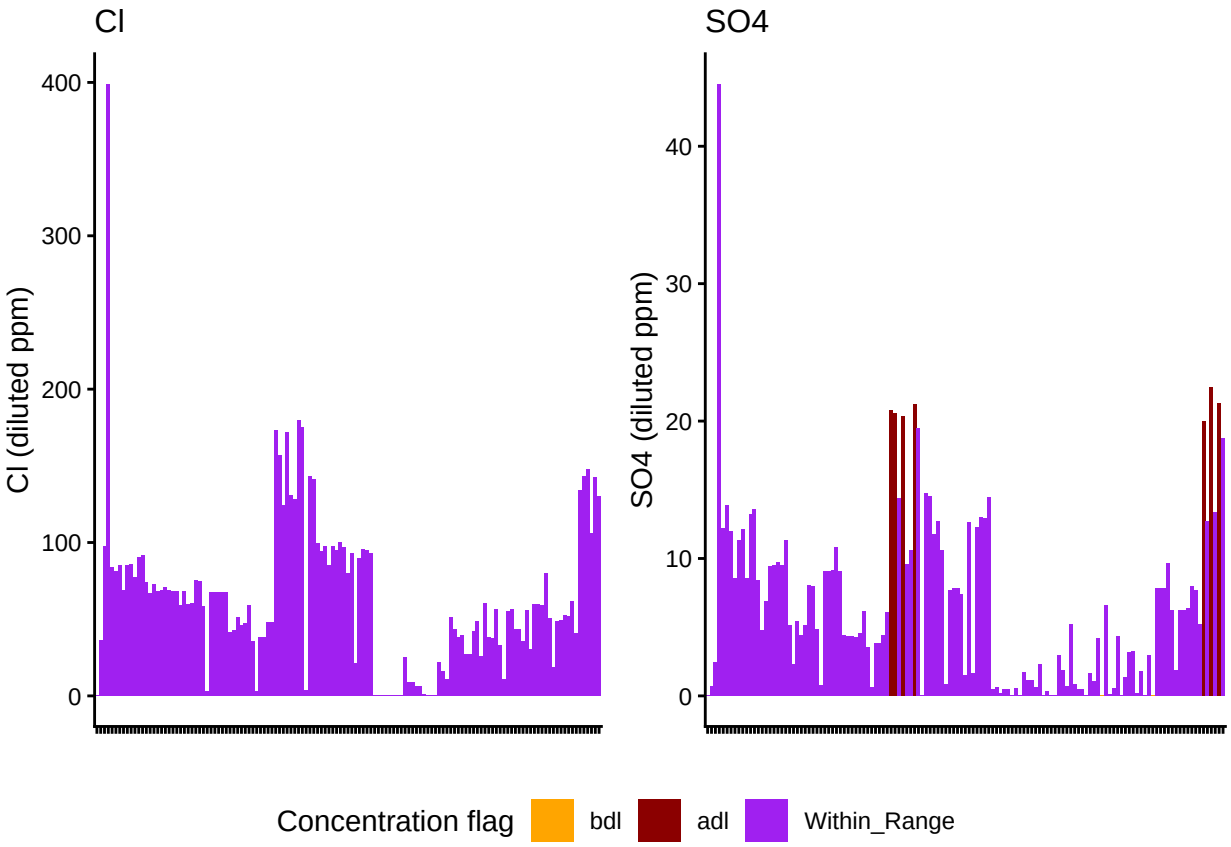
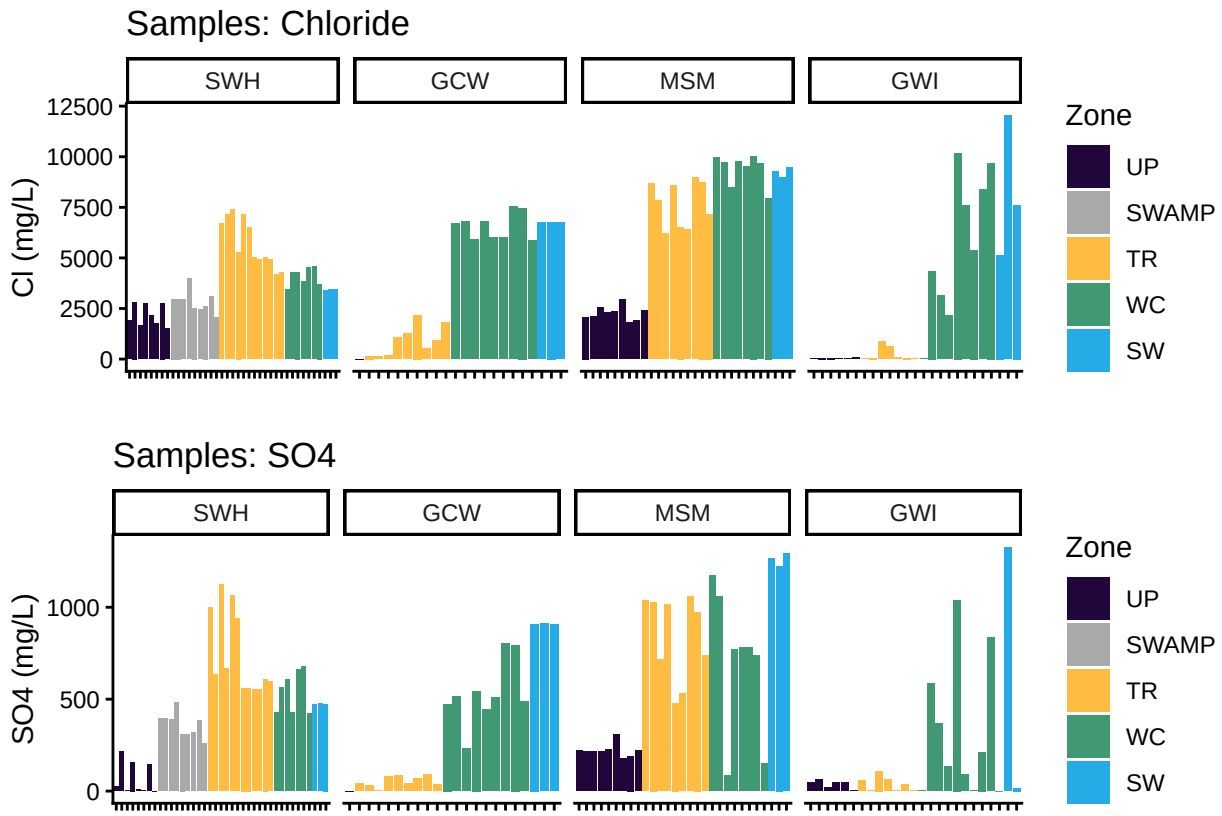


Table 6: Sample Concentration Flags

Conc_flag	Percent_samples_SO4	Percent_samples_Cl
adl	5.147059	0
bdl	1.470588	0
Within_Range	93.382353	100

1.8 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```



##Go here to compare and pull request the new log file: https://github.com/Smithsonian/BGC_QAQCtools

#end